ELE 404 – Electronic Circuits 1 **Toronto Metropolitan University**

Project Report: MOSFET Amplifier Design Project

Professor: Dr Virgilio Valente

Teaching Assistant: Shihabul Bhuiyan

Pushpinder Singh

501294986

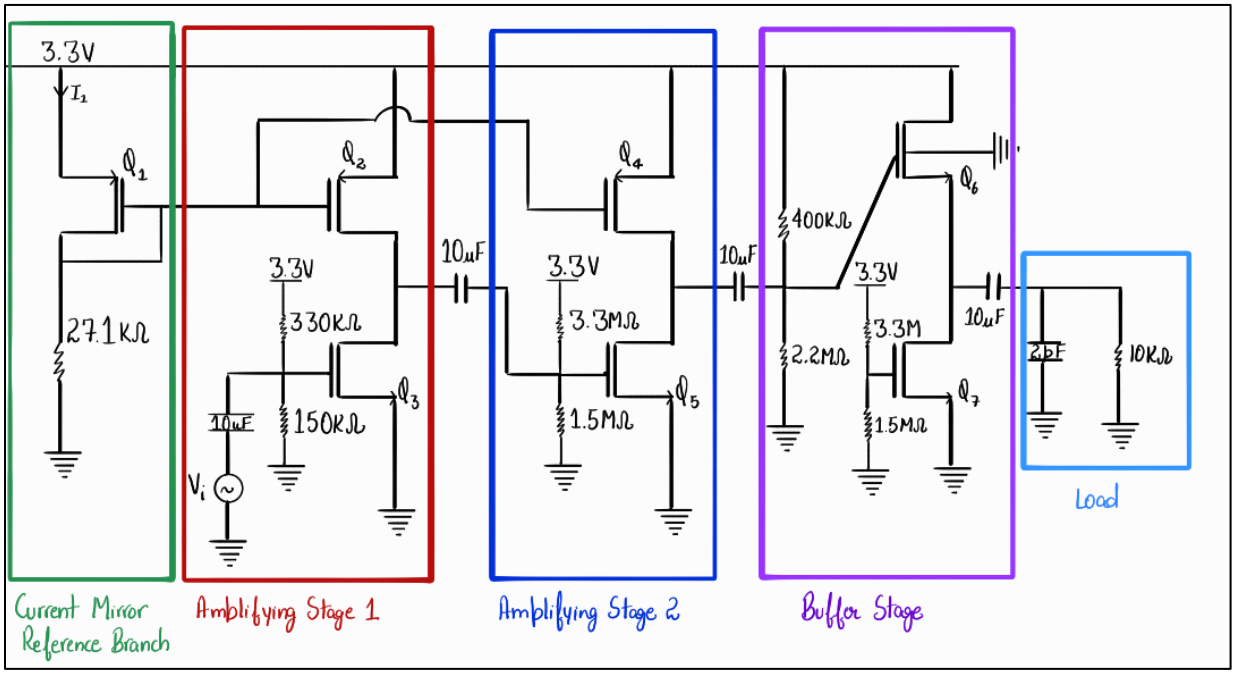
Project Objective:

The objective of this project is to design, simulate, and analyze a multi-stage CMOS amplifier that is operating on a 3.3V DC power supply and adhering to all the specifications provided in the project manual. The amplifier must drive a 10-k Ohm load and produce an output swing of 1.5V peak-peak and obtain a gain of 60 dB, while keeping the total DC power dissipation under 1mW.

Project Architecture Overview:

To achieve a high gain of 60 dB and to comply with all design specifications, a four-stage amplifier architecture was selected. The stages are isolated with 10 uF coupling capacitors to stabilize the DC operating points of all the stages. The description of all the stages used in the amplifier are given below:

1. **Current Mirror Reference Stage:** to mirror the current in both of the amplifying stages.
2. **Common Source Amplifier 1:** to amplify the input signal.
3. **Common Source Amplifier 2:** to amplify the signal from amplifier 1.
4. **NMOS Source Follower Buffer:** to prevent the drop in output when load is connected.



Current Mirror Reference Stage:

This PMOS reference branch was used to bias and mirror the reference current in both of the amplifying stages. The design consists of a diode connected PMOS and a resistor in series. This design provides high output resistance and stable biasing in the amplifying stages. The reference branch was designed to keep the current through this branch to be 80uA, which will be mirrored in the subsequent amplifying stages. This current value keeps the power consumption low for this branch as we have DC power limit of 1mW. The overdrive for this branch was chosen to be 0.4V, which lies in the middle of the overdrive specification range of 0.2V – 0.6V. By using the current formula for MOSFET in saturation, the W/L ratio turned out to be 40. Based on this ratio, the width and the length of the transistor was selected to be 20um and 0.5um respectively, which adheres to the size constraints in the project manual. Finally, to keep the branch current at 80uA and overdrive at 0.4V, a 27.1k ohm resistor was placed below the PMOS. The calculations for W/L ratio and the rating of the resistor were given below in Figure 1.0.

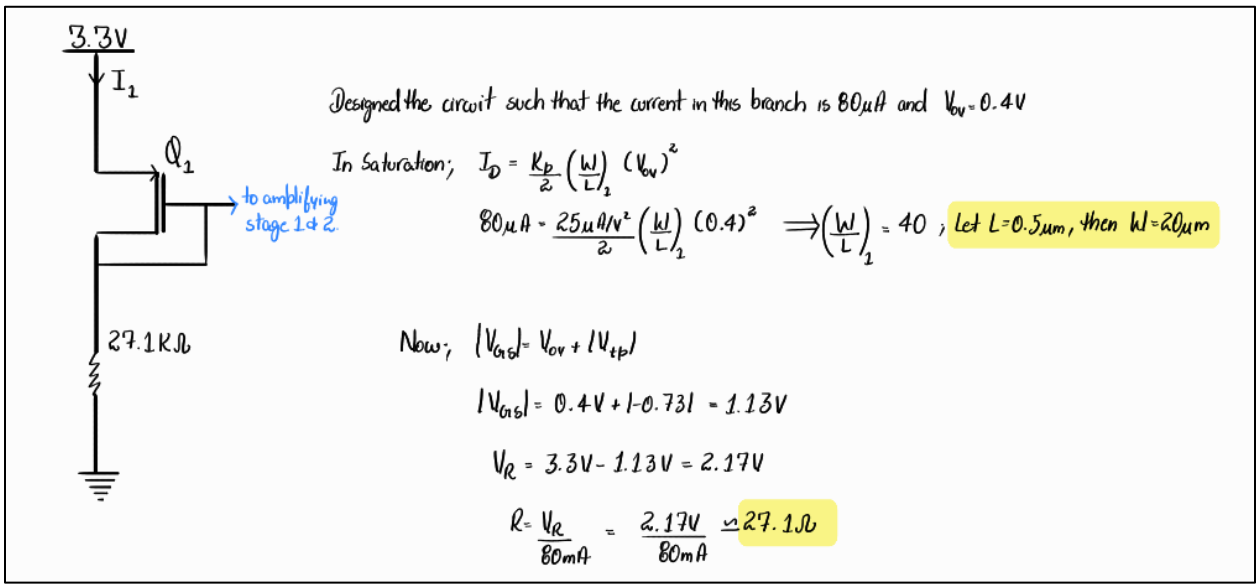


Figure 1.0: W/L and Resistor calculations for first stage

The simulations results for this stage were very close to the theoretical results, with simulation branch current being 81.1uA, which is very close to the calculated value of 80uA. Figure 1.1 shows the DC simulation results for this stage.

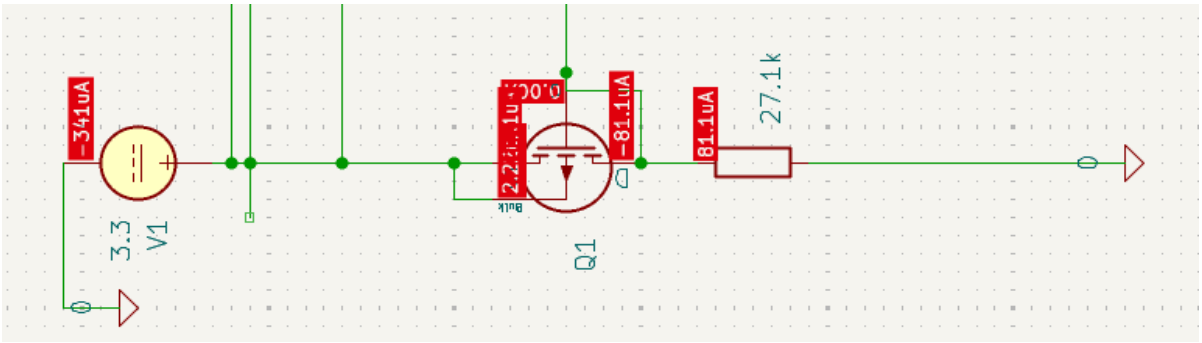


Figure 1.1: DC simulation results for stage 1.

Common Source Amplifier 1:

This is the first amplifying stage of the design consisting of PMOS(q2) and a NMOS(q3). PMOS mirrors the current from the reference stage by keeping its size and overdrive voltage to be same as the PMOS in the reference branch. However, by using the current formula and keeping the current at 80uA and overdrive at 0.4V, the NMOS(q3) had a different W/L ratio of 10. Based on this ratio, the width and length of NMOS was selected to be 5um and 0.5um, which adhered to the size constraints in the project manual. The NMOS was biased using a purely resistive voltage divider which is parallel combination of 330k and 150k resistor connected to 3.3 V DC power supply as shown in Figure 2.0. This biasing design was selected in order to maintain a Gate-Source voltage of 1.03V, which keeps the overdrive of NMOS to be 0.4V and to provide an input resistance of 103k ohm which is greater than minimum input resistance of 100k ohm specified in the project manual. This stage was connected to the subsequent amplifying stage using a coupling capacitor and a voltage divider of 3.3M ohm and 1.5M ohm resistors. These large resistors were chosen to prevent the gain of the first stage from falling as the AC gain is also dependent upon the parallel combination of these two resistors. The input signal was selected to be 0.5mV or 1V peak-peak. Due to high output resistance of PMOS and NMOS, the gain from this stage was found to be -54.43V/V or 34.71dB. The gain calculations and biasing calculations are provided in Figure 2.0 given below.

The simulations results were very close to the calculated values. The voltage at gate of NMOS (q3) was exactly 1.03 as calculated and the gain was around 35.9dB which was very close to the calculated gain. The simulation current through this branch was 83.3uA, which was slightly higher but very close to the calculated values. The reason of this small error could be channel length modulation. Figure 2.1 and 2.2 shows the simulation results of amplifying stage 1.

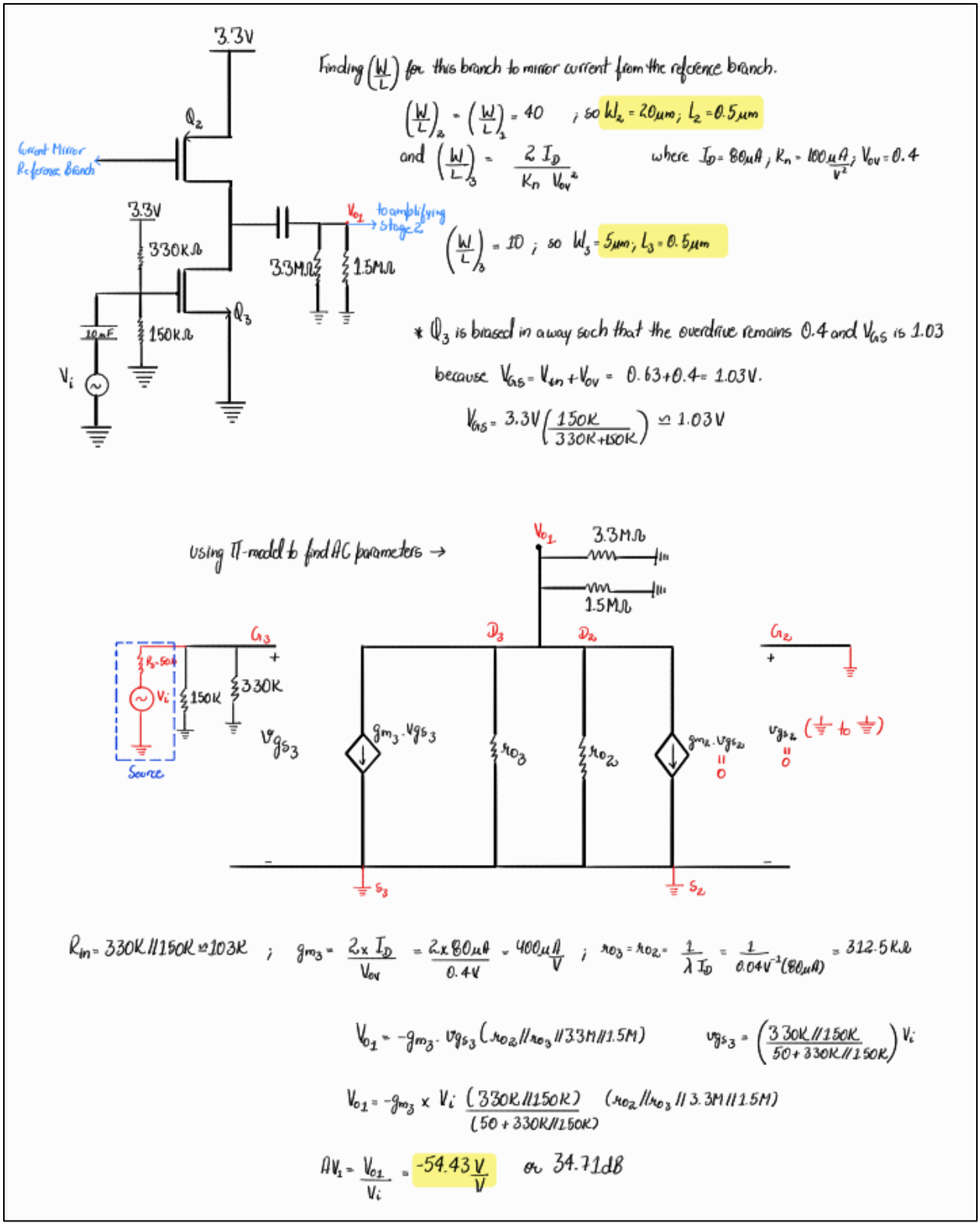


Figure 2.0: DC biasing and gain calculations for Amplifying stage 1.

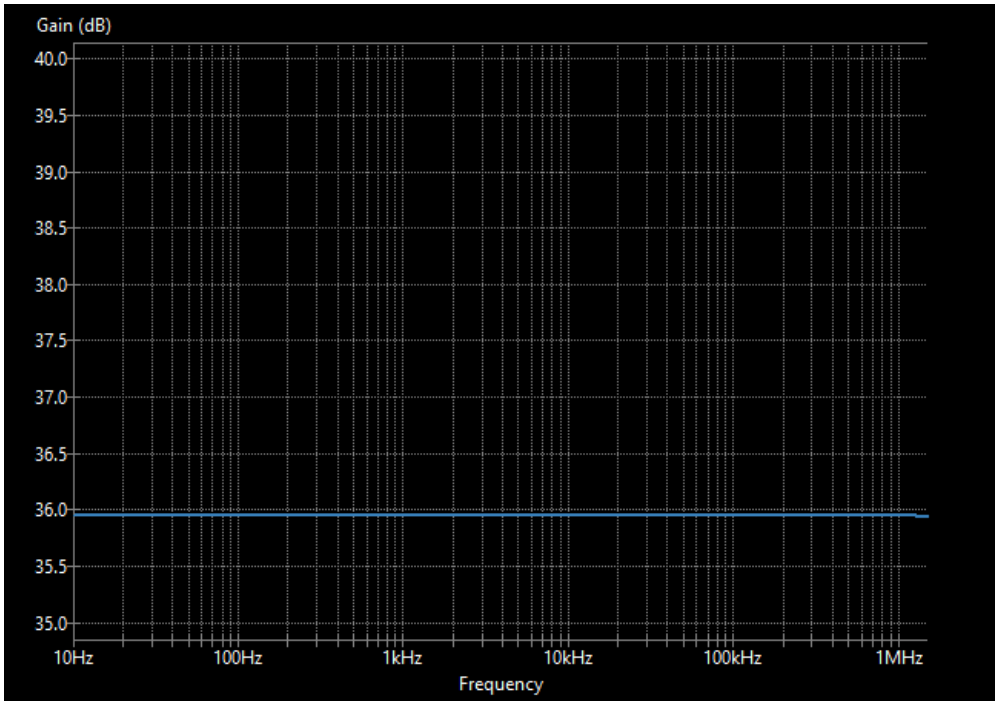


Figure 2.1: Simulation gain of Amplifying stage 1.

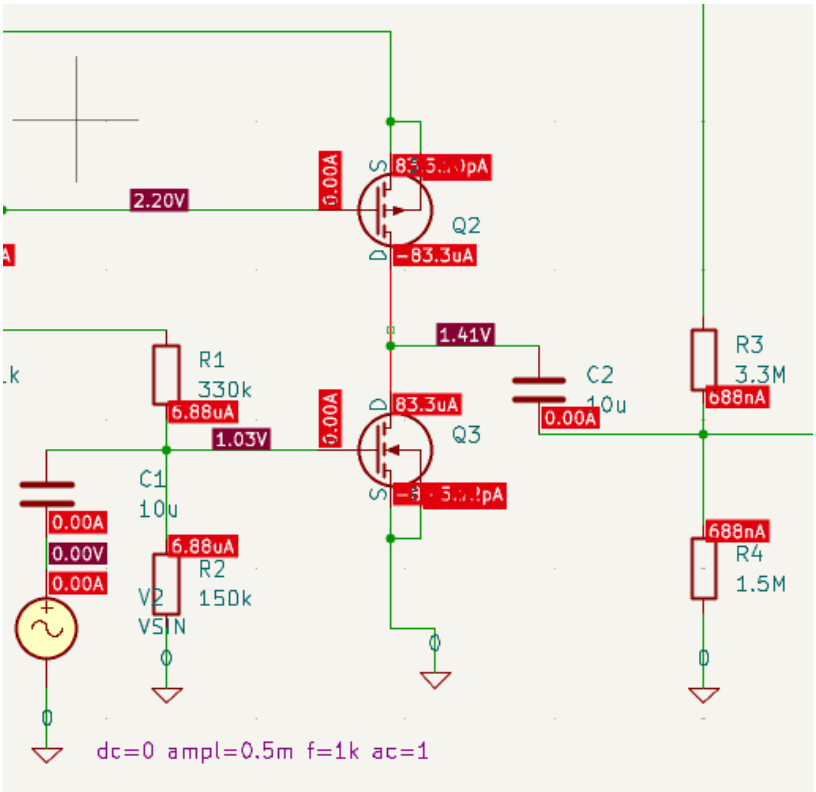


Figure 2.2: Simulation DC results of Amplifying stage 1.

Common Source Amplifier 2:

This is the second amplifying stage of the design consisting of PMOS(q4) and a NMOS(q5). Similar to the amplifying stage 1, PMOS mirrors the current from the reference stage by keeping its size and overdrive voltage to be same as the PMOS in the reference branch. However, by using the current formula and keeping the current at 80uA and overdrive at 0.4V, the NMOS had a different W/L ratio of 10. Based on this ratio, the width and length of NMOS was selected to be 5um and 0.5um, which adhered to the size constraints in the project manual. The NMOS was biased using a purely resistive voltage divider which is parallel combination of 3.3M and 1.5M resistor connected to 3.3 V DC power supply as shown in Figure 3.0. This biasing design was selected in order to maintain a Gate-Source voltage of 1.03V, which keeps the overdrive of NMOS to be 0.4V and to provide an input resistance of 1.03M ohm which is greater than minimum input resistance specified in the project manual. This stage was connected to the buffer stage using a coupling capacitor and a voltage divider of 400k ohm and 2.2M ohm resistors as shown in Figure 3.0. This high value was chosen to prevent the gain of the second stage from falling as the AC gain is also dependent upon the parallel combination of these two resistors. Originally, the specifications of the voltage divider resistors were chosen to be 1M and 2.2M, but the output signal after the buffer stage started to clip at the bottom due to the body effect. Therefore, to increase the DC resting point of the Buffer's output, the 1M ohm resistor was replaced with a 400k ohm resistor. Due to high output resistance of PMOS and NMOS, the gain from this stage was found to be -50.43V/V or 32.62dB. The gain calculations and biasing calculations are provided in Figure 3.0 given below.

The simulation results closely follow the results from theoretical calculations. The current through this branch was 83.3uA. The voltage at the gate of NMOS(q5) was exactly equal to 1.03V as calculated and the gain from this stage was 33.7dB, which is very close to the calculated gain of 32.62dB. The simulation results ae shown in Figure 3.1 and 3.2.

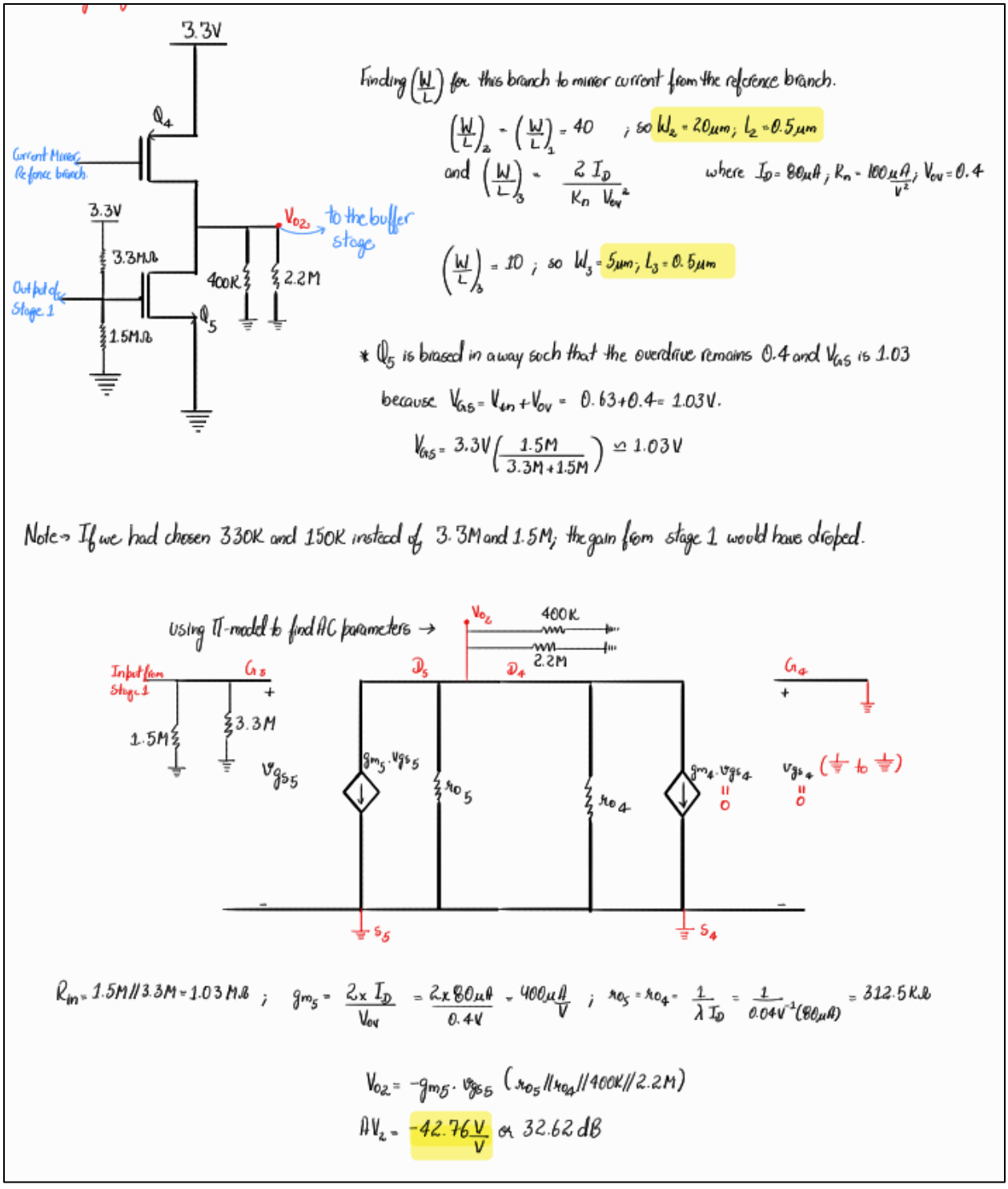


Figure 3.0: DC biasing and gain calculations for Amplifying stage 2.

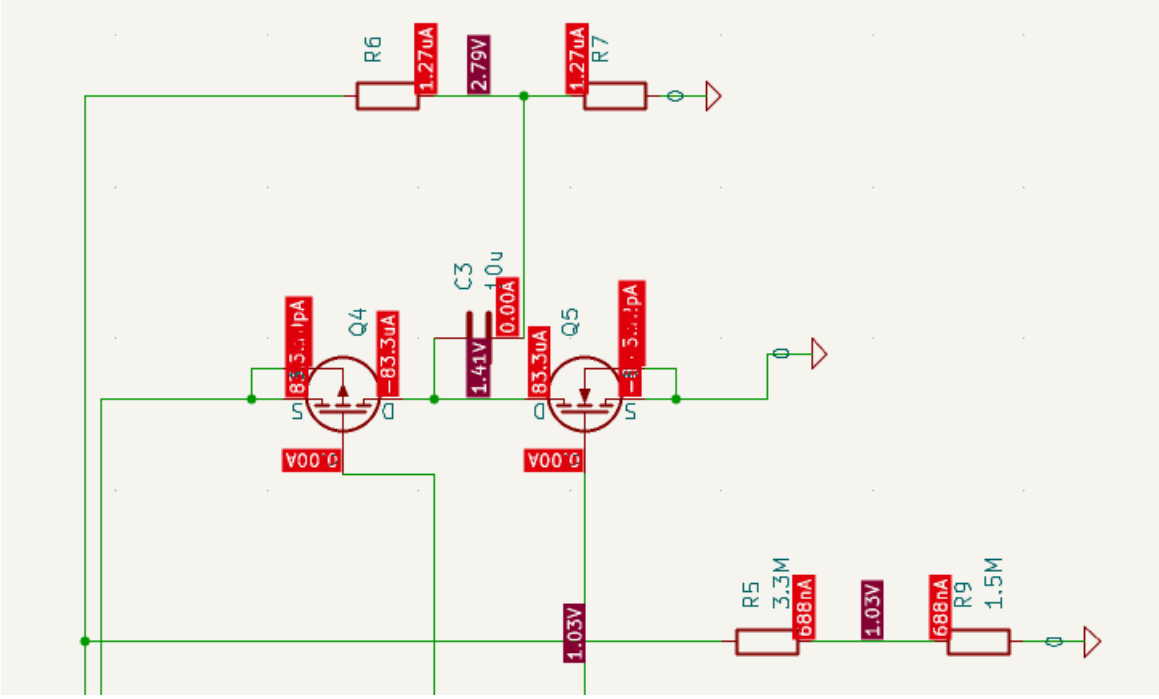


Figure 3.1: Simulation DC results of Amplifying stage 2.

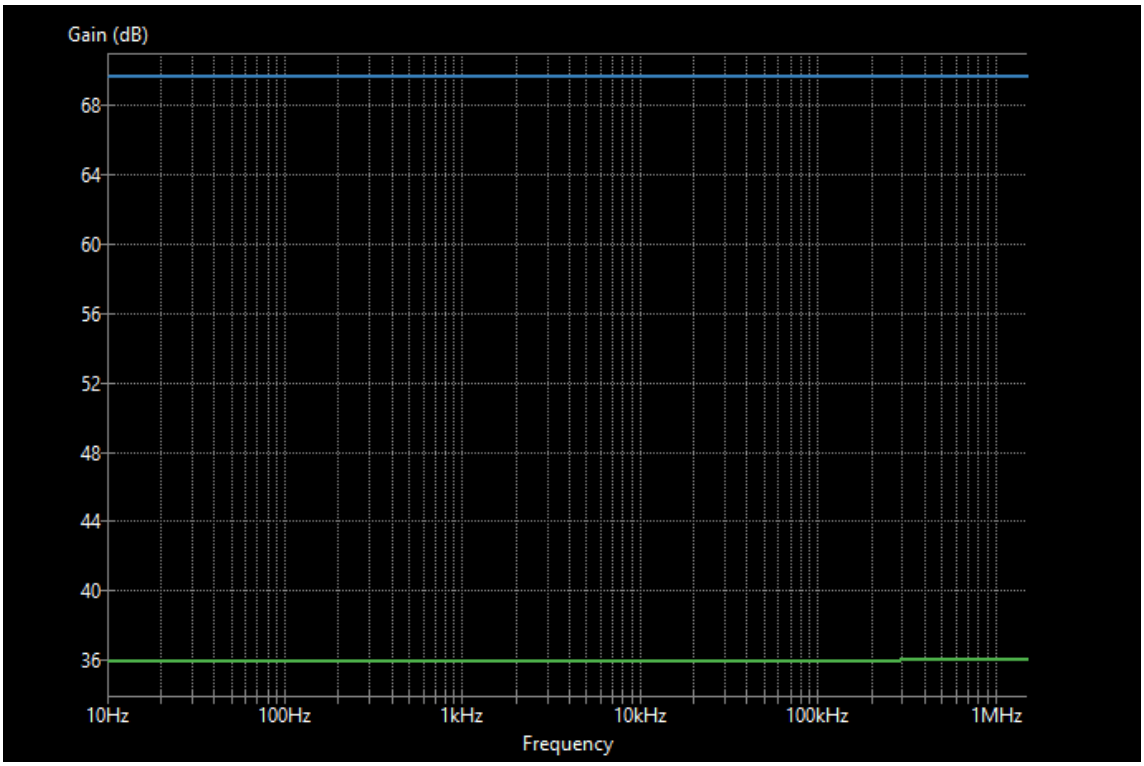


Figure 3.2: Simulation gain after the Amplifying stage 1 (green) & Amplifying stage 2 (blue).

NMOS Source Follower Buffer:

To prevent the gain from dropping, a buffer stage was added to the architecture of the amplifier. A NMOS(q6) source follower was used as a buffer with its bulk connected to the ground and the source connected to drain of another NMOS (q7). The transistor q7 is biased using a resistive voltage divider with 3.3M ohm and 1.5M ohm resistors in parallel combination as shown in Figure 4.0. The current though this branch was kept to 80uA by biasing q7 with 1.03V Gate-Source voltage and a W/L ratio of 10. Therefore, by using the current formula, the current through this branch was calculated to be 80uA. The transistor q6 was biased with another voltage divider with 400k ohm and 2.2M ohm resistors in parallel combination as shown in Figure 4.0. This biasing arrangement keeps the voltage at the gate of q6 at around 2.79V. In an ideal case, where q6 does not suffer with body effect, the DC resting point at the output would be around 1.76 V, but due to the body effect the threshold voltage of the q6 rises by 0.35 V. Therefore, the non-ideal DC resting point at the output becomes around 1.41V. The calculated gain which includes the body effect calculations was around -1.62dB. The total gain after this stage was 65.71dB. The DC and AC gain calculations for this stage are shown in Figure 4.0.

The simulation results aligned with the theoretical results with the current being 83.3uA and the DC resting point at the output to be 1.40V which is very close to the theoretical value of 1.41V. The simulated AC gain from this stage was -1.7dB which is close to the calculated gain of -1.62dB from this stage. The simulation DC values and AC gain values are given in Figure 4.1 and 4.2.

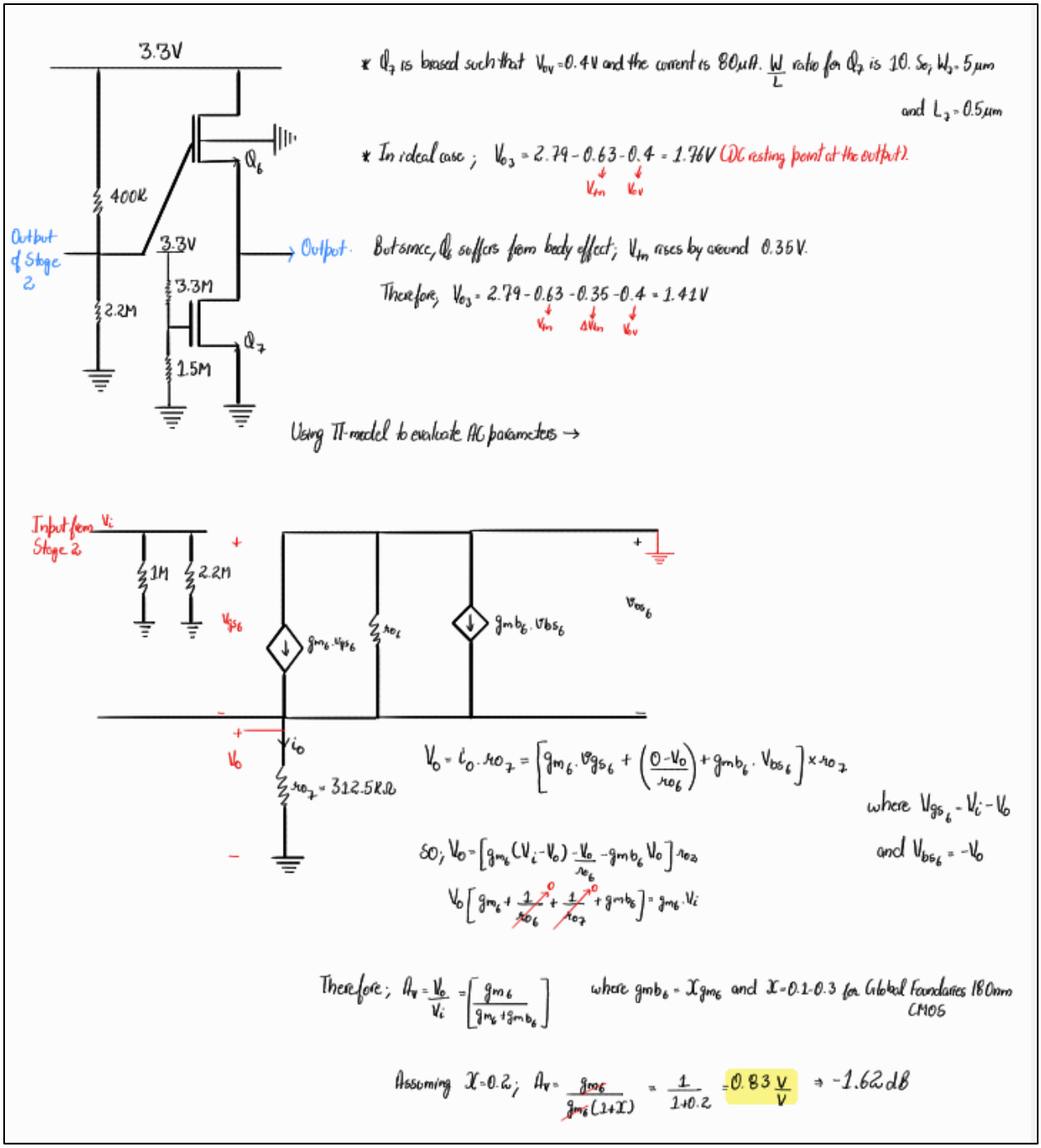


Figure 4.0: DC biasing and gain calculations for Buffer stage.

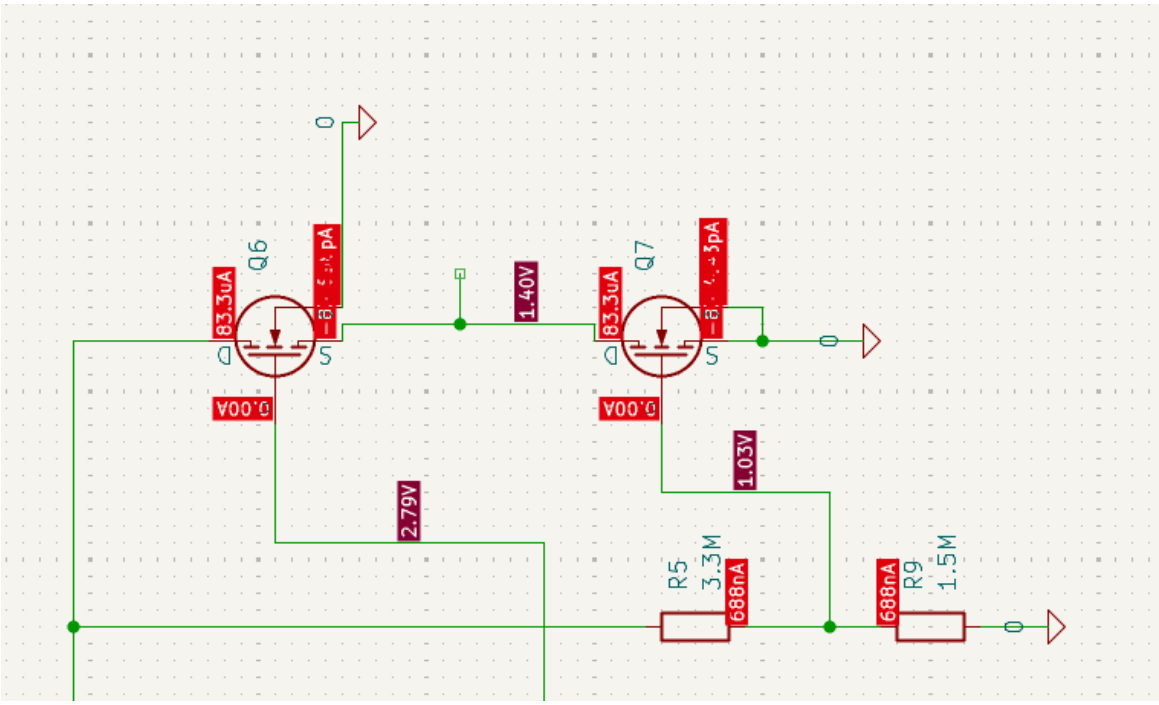


Figure 4.1: Simulation DC results of the Buffer Stage.

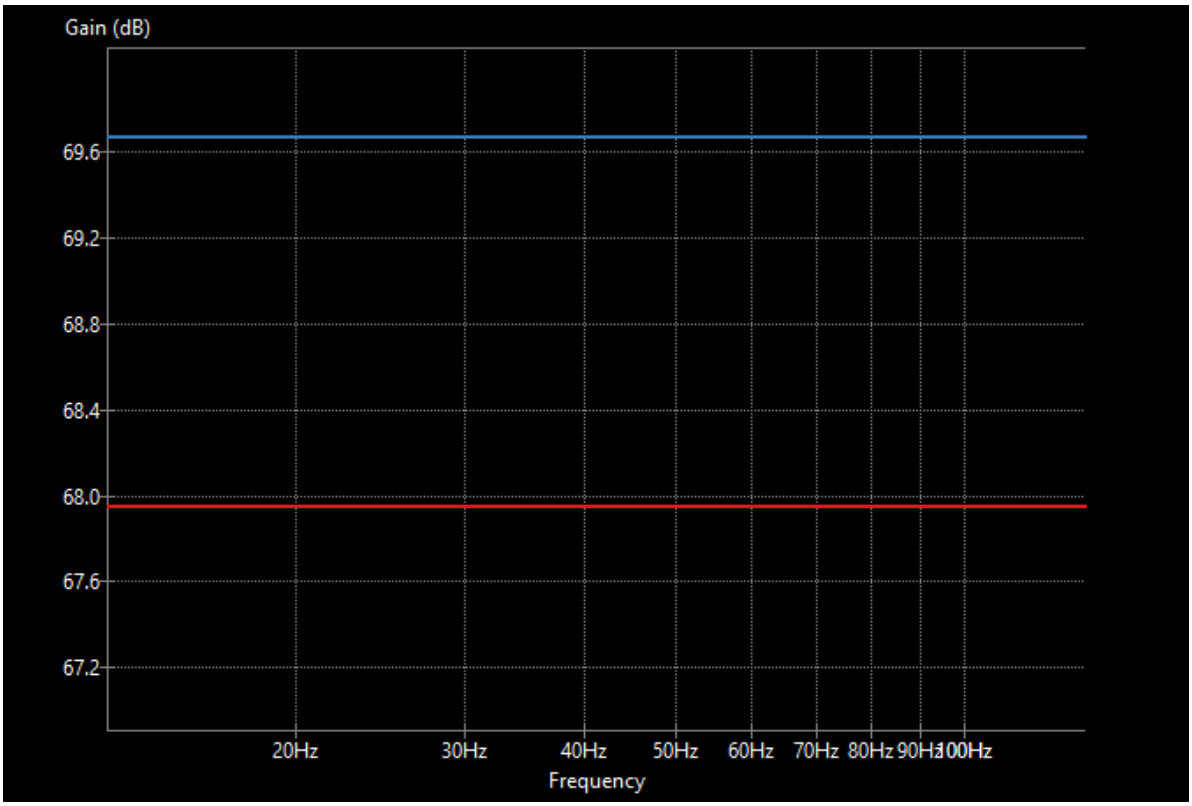


Figure 4.2: Simulation gain after the buffer stage (red) & Amplifying stage 2 (blue).

Output:

The load resistor of 10k ohm was connected to the output of the buffer stage with a load capacitance of 10 pF. The load was connected to the buffer using a 10uF coupling capacitor. The no-load gain this amplifier architecture was 67.9dB and the gain with the load resistor is 66.75dB, which is only 1.15 dB lower than no-load gain. Hence, the load sensitivity of the amplifier is about 1.7% which is well under the 10% load sensitivity limit. Figure 5.0 shows the overall gain of the amplifier.

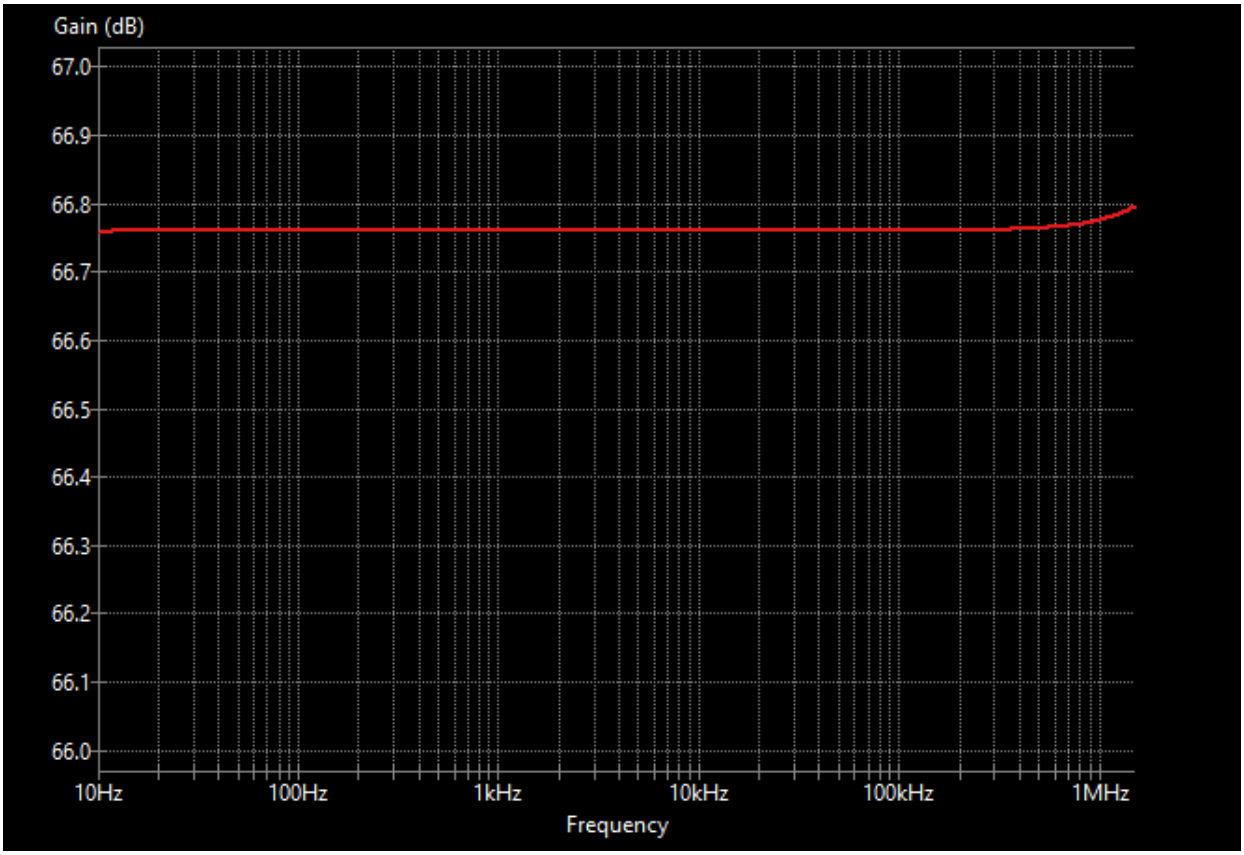


Figure 5.0: Overall gain of the amplifier

The output waveform ranges from 0.48V to 2.2V producing an output swing of 1.72 peak-peak which is higher than the project's objective. Figure 6.0 shows the output waveform of the amplifier.

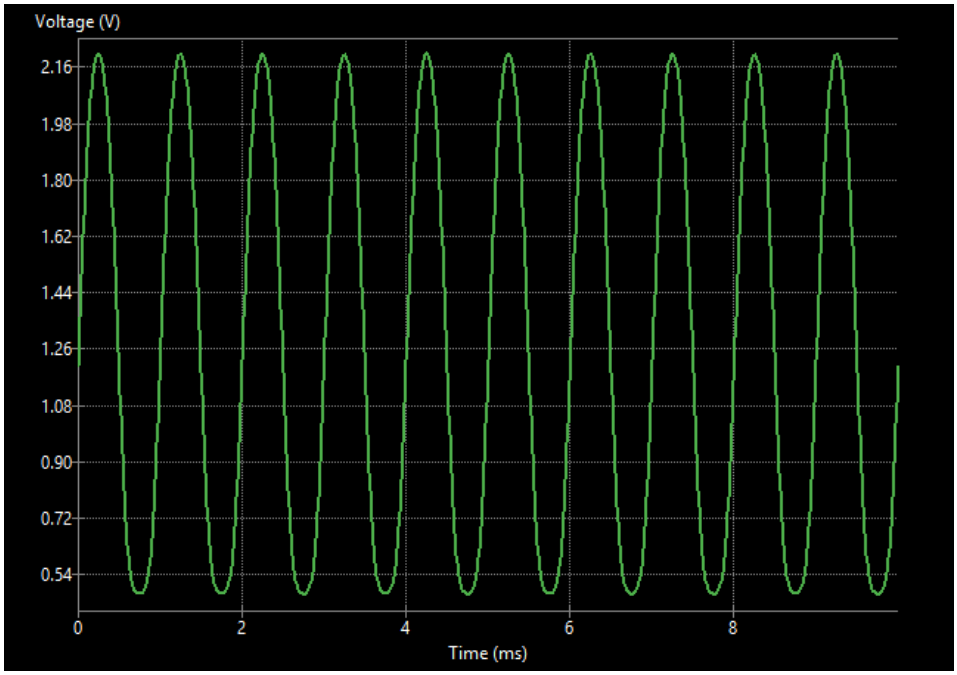


Figure 6.0: Output signal waveform of the amplifier.

The power dissipation of the whole amplifier is calculated as follows:

$$P_{diss} = DC\ Power\ Supply\ \times sum\ of\ all\ currents\ in\ all\ stages$$

$$P_{diss} = 3.3V\ \times\ (331uA) = 1.09\ mW$$

The power dissipation is slightly higher than the 1mW limit specified. This is due to high amounts to current flowing through each branch, however if we reduce the current through each branch, the power dissipation gets lower, but the transconductance gets lower and hence the gain decreases, and it is very difficult to obtain an output swing of 1.5V peak-peak. Therefore, we must accept this trade off in order to obtain a high gain and 1.5V peak-peak output swing.